

Homework 3

Zhiqi Gan

Problem 3.1. (Dimensions in matrix multiplications) Say A, B, C are $3 \times 5, 5 \times 3, 3 \times 1$ matrices respectively, then which of $BA, AB, ABAB, BAC, BABC$ are well-defined?

Solution 3.1.

All of them are well-defined except BAC .

Problem 3.2. Calculate the following matrix multiplications. Or maybe it is undefined!

Solution 3.2.

1. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$. Not defined.

2. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 9 & 5 \\ 12 & 21 & 13 \end{bmatrix}$.

3. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Not defined.

4. $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}$.

5. $\begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$.

6. $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. Not defined.

7. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$.

8. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$.

9. $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 11 \\ 18 \end{bmatrix}$.

10. $\begin{bmatrix} 1 & 2 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} x^2 \\ x \\ 1 \end{bmatrix} = \begin{bmatrix} (x+1)^2 \\ x+1 \\ 1 \end{bmatrix}$.

11. $\begin{bmatrix} 1 & 2 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} (x+1)^2 \\ x+1 \\ 1 \end{bmatrix} = \begin{bmatrix} (x+2)^2 \\ x+2 \\ 1 \end{bmatrix}$.

12. $\begin{bmatrix} 1 & 2 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} (x+2)^2 \\ x+2 \\ 1 \end{bmatrix} = \begin{bmatrix} (x+3)^2 \\ x+3 \\ 1 \end{bmatrix}$. Pattern: $\begin{bmatrix} 1 & 2 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix}^k \begin{bmatrix} x^2 \\ x \\ 1 \end{bmatrix} = \begin{bmatrix} (x+k)^2 \\ x+k \\ 1 \end{bmatrix}, k = 1, 2, \dots$

Problem 3.3. Calculate the following matrix multiplications.

Solution 3.3.

$$1. \begin{bmatrix} 1 & 1 & 1 \\ & 1 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ 1 & 1 & \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

$n \times n$ pattern:

$$\begin{bmatrix} 1 & \cdots & 1 \\ & \ddots & \vdots \\ & & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ \vdots & \ddots & \\ 1 & \cdots & 1 \end{bmatrix} = \begin{bmatrix} n & n-1 & \cdots & 1 \\ n-1 & n-1 & \cdots & 1 \\ \vdots & & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}$$

$$2. \begin{bmatrix} & & 1 \\ & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & \\ 1 & & \end{bmatrix} = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ 3 & 2 & 1 \end{bmatrix}.$$

$n \times n$ pattern:

$$\begin{bmatrix} 1 & & & \\ 2 & \ddots & & \\ \vdots & \ddots & \ddots & \\ n & \cdots & 2 & 1 \end{bmatrix}$$

$$3. \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}^2 = 3 \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}.$$

$n \times n$ pattern:

$$\text{Set } A = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}. \text{ And all columns of } A \text{ are } \mathbf{v} \text{ itself, therefore,}$$

$$A = \mathbf{v} [1 \quad 1 \quad \cdots \quad 1] = \mathbf{v} \mathbf{v}^T$$

So,

$$A^2 = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}^2 = \mathbf{v} \mathbf{v}^T \mathbf{v} \mathbf{v}^T = \mathbf{v} (\mathbf{v}^T \mathbf{v}) \mathbf{v}^T = n \mathbf{v} \mathbf{v}^T = nA = n \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}.$$

$$4. \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}^2 = \begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}.$$

$n \times n$ pattern:

Note that

$$\begin{bmatrix} 1 & -1 & \cdots & -1 \\ -1 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & & & \\ & 2 & & \\ & & \ddots & \\ & & & 2 \end{bmatrix} - \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix}.$$

Hence,

$$\begin{aligned}
\begin{bmatrix} 1 & -1 & \cdots & -1 \\ -1 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & 1 \end{bmatrix}^2 &= \left(\begin{bmatrix} 2 & & & \\ & 2 & & \\ & & \ddots & \\ & & & 2 \end{bmatrix} - \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} \right)^2 \\
&= \begin{bmatrix} 4 & & & \\ & 4 & & \\ & & \ddots & \\ & & & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} + n \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} \\
&= \begin{bmatrix} 4 & & & \\ & 4 & & \\ & & \ddots & \\ & & & 4 \end{bmatrix} + (n-4) \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{bmatrix} \\
&= \begin{bmatrix} n & n-4 & \cdots & n-4 \\ n-4 & n & \ddots & \vdots \\ \vdots & \ddots & \ddots & n-4 \\ n-4 & \cdots & n-4 & n \end{bmatrix}
\end{aligned}$$

$$5. \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}^2 = 3 \begin{bmatrix} 1 & -1 & 1 \\ -1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix}.$$

$n \times n$ pattern:

Method 1: (This solution is from Mikaela Wuisan, Zijing College)

Set the matrix is A , let $\mathbf{v} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \\ \vdots \\ (-1)^n \end{bmatrix}$. And by observation,

$$A = \mathbf{v} \begin{bmatrix} 1 & -1 & 1 & -1 & \cdots & (-1)^n \end{bmatrix} = \mathbf{v} \mathbf{v}^T$$

So $A^2 = \mathbf{v} \mathbf{v}^T \mathbf{v} \mathbf{v}^T = \mathbf{v} (\mathbf{v}^T \mathbf{v}) \mathbf{v}^T = n \mathbf{v} \mathbf{v}^T = nA$.

Method 2:

Set the matrix is A , and $B = A^2$, then the i, j entry of A is $A_{ij} = (-1)^{i+j}$. By the law of matrix multiplication, the i, j entry of B is

$$B_{ij} = \sum_{k=1}^n A_{ik} A_{kj} = \sum_{k=1}^n (-1)^{i+k} (-1)^{k+j} = \sum_{k=1}^n (-1)^{2k} (-1)^{i+j} = \sum_{k=1}^n (-1)^{i+j} = n(-1)^{i+j} = nA_{ij}$$

That is to say, every entry of B is n times the same entry of A , so $A^2 = nA$.

Problem 3.4. Calculate the following matrix multiplications.

1. Suppose $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are mutually orthogonal unit vectors in $\mathbb{R}^5$. Compute

$$\begin{bmatrix} \mathbf{u}^T \\ \mathbf{v}^T \\ \mathbf{w}^T \end{bmatrix} \begin{bmatrix} \mathbf{u} & \mathbf{v} & \mathbf{w} \end{bmatrix}.$$

2. Compute $\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix}^2$. (This is actually the square of a reflection.)

3. $\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & \sin \phi \\ \sin \phi & -\cos \phi \end{bmatrix}$. This is a product of two reflections, and the result is a rotation. What is the angle of this rotation? (Can you also find a geometric proof?)

Solution 3.4.

1. $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are mutually orthogonal unit vectors $\implies \mathbf{u}^T \mathbf{u} = \mathbf{v}^T \mathbf{v} = \mathbf{w}^T \mathbf{w} = 1$ and the dot product between them is

0. Therefore, $\begin{bmatrix} \mathbf{u}^T \\ \mathbf{v}^T \\ \mathbf{w}^T \end{bmatrix} \begin{bmatrix} \mathbf{u} & \mathbf{v} & \mathbf{w} \end{bmatrix} = I_3$.

2. I_2 .

- 3.

$$\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{bmatrix} \begin{bmatrix} \cos \phi & \sin \phi \\ \sin \phi & -\cos \phi \end{bmatrix} = \begin{bmatrix} \cos \theta \cos \phi + \sin \theta \sin \phi & \cos \theta \sin \phi - \sin \theta \cos \phi \\ \sin \theta \cos \phi - \cos \theta \sin \phi & \sin \theta \sin \phi + \cos \theta \cos \phi \end{bmatrix} = \begin{bmatrix} \cos(\theta - \phi) & -\sin(\theta - \phi) \\ \sin(\theta - \phi) & \cos(\theta - \phi) \end{bmatrix}$$

The angle of this rotation is $(\theta - \phi)$.

Problem 3.5. Calculate the following matrix powers.

1. $\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^2$.

2. $\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^3$.

3. $\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^{2020}$.

4. $\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^{-1}$.

5. $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}^2$.

6. $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}^3$.

7. $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}^{2020}$.

8. $\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^2$.

9. $\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^3$.

10. $\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^{2020}$.

$$11. \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^{-1}.$$

Solution 3.5.

1. I_2 .

$$2. \text{ From 1, } \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^3 = \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^2 \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix} = I_2 \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix} = \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}.$$

$$3. \text{ From 1, } \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^{2020} = \left(\begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^2 \right)^{1010} = I_2.$$

$$4. \text{ Since the inverse is unique, from 1, we immediately have } \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}^{-1} = \begin{bmatrix} 11 & 6 \\ -20 & -11 \end{bmatrix}.$$

$$5. \text{ Set } A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}. \text{ By observation, } A = \mathbf{v}\mathbf{v}^T.$$

So

$$A^2 = \mathbf{v}\mathbf{v}^T\mathbf{v}\mathbf{v}^T = \mathbf{v}(\mathbf{v}^T\mathbf{v})\mathbf{v}^T = 4\mathbf{v}\mathbf{v}^T = 4A = 4 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$6. \text{ Similar to 5, Set } A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}. \text{ And we also have } A = \mathbf{v}\mathbf{v}^T. \text{ So}$$

$$A^3 = \mathbf{v}\mathbf{v}^T\mathbf{v}\mathbf{v}^T\mathbf{v}\mathbf{v}^T = \mathbf{v}(\mathbf{v}^T\mathbf{v})(\mathbf{v}^T\mathbf{v})\mathbf{v}^T = 4^2\mathbf{v}\mathbf{v}^T = 16A = 16 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

7. Similar to 5 and 6,

$$A^{2020} = (\mathbf{v}\mathbf{v}^T)^{2020} = \mathbf{v}(\mathbf{v}^T\mathbf{v})^{2019}\mathbf{v}^T = 4^{2019}A = 4^{2019} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

8.

$$\text{Note that } \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & & & \\ & 2 & & \\ & & 2 & \\ & & & 2 \end{bmatrix} - \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}. \text{ Thus,}$$

$$\begin{aligned} \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & 1 \end{bmatrix}^2 &= \left(\begin{bmatrix} 2 & & & \\ & 2 & & \\ & & 2 & \\ & & & 2 \end{bmatrix} - \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \right)^2 \\ &= \begin{bmatrix} 4 & & & \\ & 4 & & \\ & & 4 & \\ & & & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} - 2 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} + 4 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4 & & & \\ & 4 & & \\ & & 4 & \\ & & & 4 \end{bmatrix} \end{aligned}$$

9. From 8, we have

$$\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{bmatrix}^3 = \begin{bmatrix} 4 & & & \\ & 4 & & \\ & & 4 & \\ & & & 4 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{bmatrix} = 4 \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{bmatrix}$$

10.

$$\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{bmatrix}^{2020} = \begin{bmatrix} 4 & & & \\ & 4 & & \\ & & 4 & \\ & & & 4 \end{bmatrix}^{1010} = 4^{1010} I_4.$$

11.

From 8, it's obvious to see that

$$\begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{bmatrix}^{-1} = \frac{1}{4} \begin{bmatrix} 1 & -1 & -1 & -1 \\ -1 & 1 & -1 & -1 \\ -1 & -1 & 1 & -1 \\ -1 & -1 & -1 & -1 \end{bmatrix}.$$